



भारतीय प्रौद्योगिकी संस्थान मुंबई

Indian Institute of Technology Bombay

CS 6001: Game Theory and Algorithmic Mechanism Design

Week 6

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Slide preparation acknowledgments: Ramsundar Anandanarayanan and Harshvardhan Agarwal

ज्ञानम् परमम् ध्येयम्

Knowledge is the supreme goal



- ▶ Equilibrium in IIEFGs
- ▶ Game Theory in Practice: P2P File Sharing
- ▶ Bayesian Games
- ▶ Strategy, Utility in Bayesian Games
- ▶ Equilibrium in Bayesian Games
- ▶ Examples in Bayesian Equilibrium



- Can extend the subgame perfection of PIEFG, but since the nodes/histories are uncertain, we need to extend to mixed strategies
- Because of the information sets, best response cannot be defined without the belief of each player

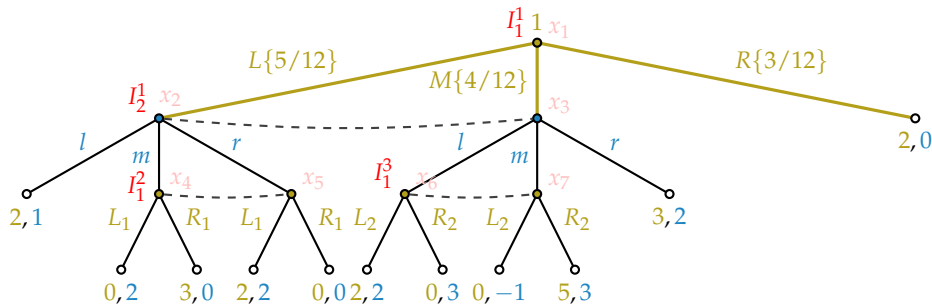
Belief

It is the conditional probability distribution over the histories in an information set - conditioned on reaching the information set.



Example: An IIEFG with perfect recall

EX 7.38 MSZ: An IIEFG with perfect recall, i.e., mixed and behavioral strategies are equivalent.

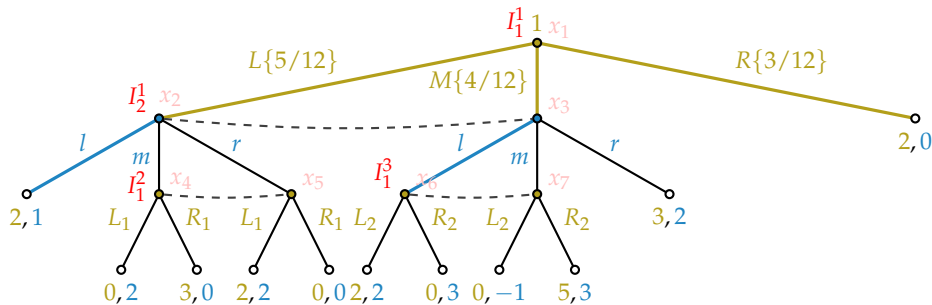


Consider the behavioral strategy profile: σ_1 , at $I_1^1(L\{5/12\}, M\{4/12\}, R\{3/12\})$



Example: An IIEFG with perfect recall

EX 7.38 MSZ: An IIEFG with perfect recall, i.e., mixed and behavioral strategies are equivalent.

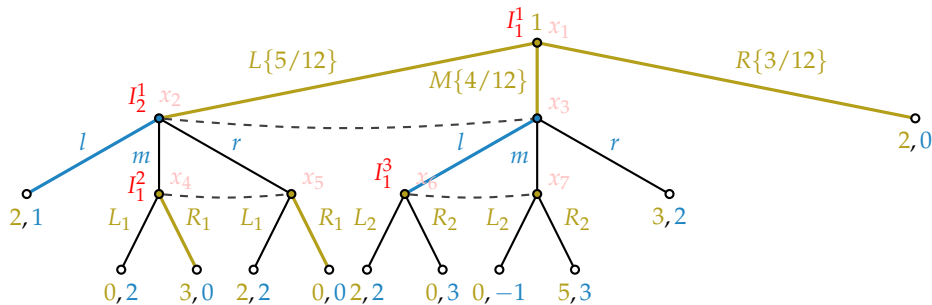


Consider the behavioral strategy profile: σ_2 , at $I_2^1(l\{1\}, m\{0\}, r\{0\})$ choose l



Example: An IIEFG with perfect recall

EX 7.38 MSZ: An IIEFG with perfect recall, i.e., mixed and behavioral strategies are equivalent.

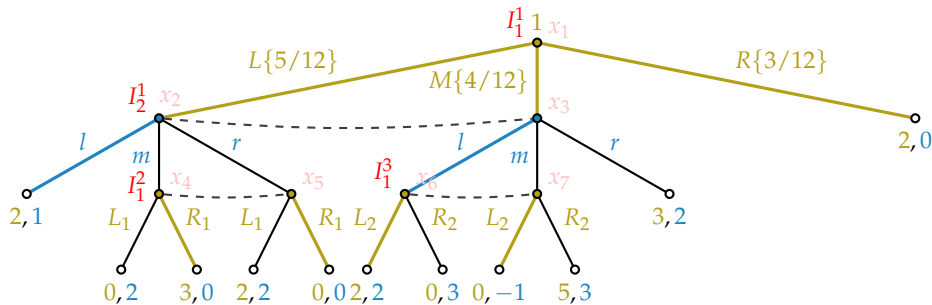


Consider the behavioral strategy profile: σ_1 , at $I_1^2(L_1\{0\}, R_1\{1\})$ choose R_1



Example: An IIEFG with perfect recall

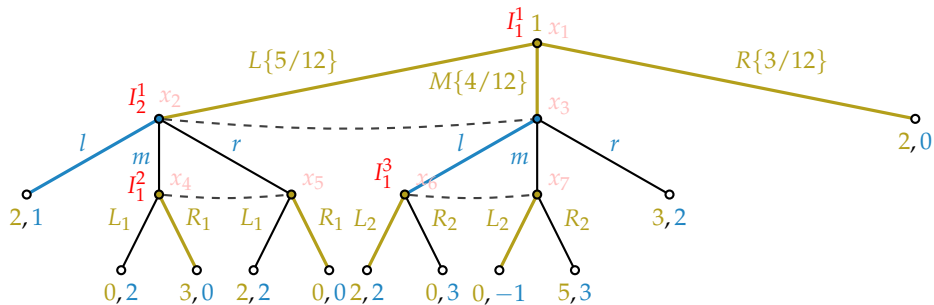
EX 7.38 MSZ: An IIEFG with perfect recall, i.e., mixed and behavioral strategies are equivalent.



Consider the behavioral strategy profile: σ_1 , at $I_1^3(L_2\{1\}, R_2\{0\})$ choose L_2



Example: An IIEFG with perfect recall



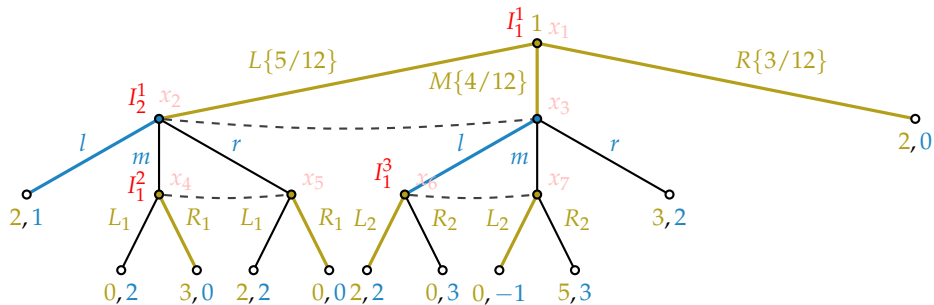
Question

Is this an equilibrium?
which implies

- Are the Bayesian beliefs consistent with P_σ - that visits vertex x with probability $P_\sigma(x)$?
- The actions and beliefs are consistent for every player, i.e., maximizes their expected utility?



Example: An IIEFG with perfect recall



Sequential rationality

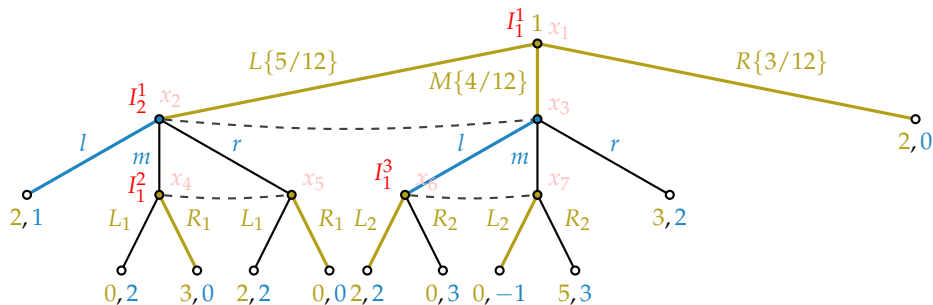
Choose an action maximizing expected utility at each information set.

The strategy vector σ induces the following probabilities to the vertices.

$$P_{\sigma}(x_2) = 5/12, P_{\sigma}(x_3) = 4/12, P_{\sigma}(x_4) = 0, P_{\sigma}(x_5) = 0, P_{\sigma}(x_6) = 4/12, P_{\sigma}(x_7) = 0$$



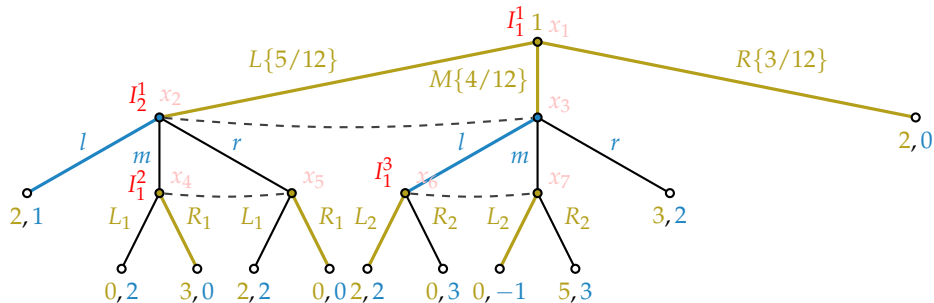
Example: An IIEFG with perfect recall



- Player 1 at information set I_1^3 , believes that x_6 is reached with probability 1.
- If the belief was $> 2/7$ in favor of x_7 , player 1 should have chosen R_2



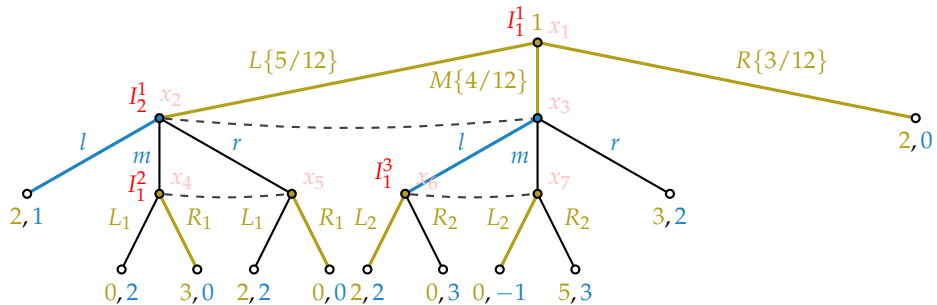
Example: An IIEFG with perfect recall



- Player 2 at I_2^1 believes the x_3 is reached w.p. $P_\sigma(x_3|I_2^1) = P_\sigma(x_3)/(P_\sigma(x_2) + P_\sigma(x_3)) = 4/9$
- Similarly $P_\sigma(x_2|I_2^1) = 5/9$



Example: An IIEFG with perfect recall



Question

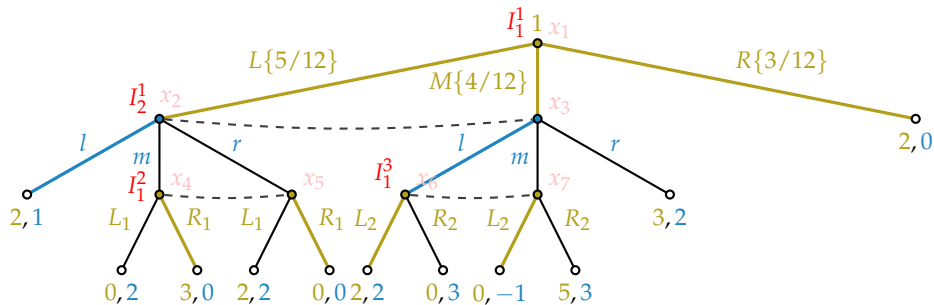
Is the action of player 2 sequentially rational w.r.t. her belief?

Answer

By picking l , expected utility $= 5/9 \times 1 + 4/9 \times 2 = 13/9$, larger than any other choice of action.



Example: An IIEFG with perfect recall



Question

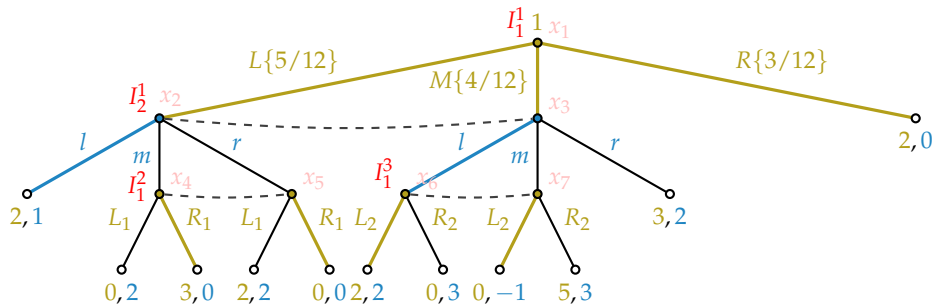
Given all information, what is the sequentially rational strategy for player 1 at I_1^1

Answer

L, M, R all give the same expected utility for player 1 (utility = 2).



Example: An IIEFG with perfect recall



Thus, mixed/behavioral strategy profile σ is sequentially rational for all players.



Belief

Let the information sets of player i be $I_i = \{I_i^1, I_i^2, I_i^3, \dots, I_i^{k(i)}\}$.

The belief of player i is a mapping $\mu_i^j : I_i^j \rightarrow [0, 1]$ s.t., $\sum_{x \in I_i^j} \mu_i^j(x) = 1$

Bayesian belief

A belief $\mu_i = \{\mu_i^1, \mu_i^2, \dots, \mu_i^{k(i)}\}$ of player i is Bayesian w.r.t.to the behavioral strategy σ , if it is derived from σ using Bayes rule, i.e.,

$$\mu_i^j(x) = P_\sigma(x) / \sum_{y \in I_i^j} P_\sigma(y), \forall x \in I_i^j, \forall j = 1, 2, 3, \dots, k(i)$$



Sequential rationality

A strategy σ_i of player i at an information set I_i^j is sequentially rational given σ_{-i} and partial belief μ_i^j if

$$\sum_{x \in I_i^j} \mu_i^j(x) u_i(\sigma_i, \sigma_{-i} | x) \geq \sum_{x \in I_i^j} \mu_i^j(x) u_i(\sigma'_i, \sigma_{-i} | x)$$

- The tuple (σ, μ) is sequentially rational if it is sequentially rational for every player at every information set.
- The tuple (σ, μ) is also called an assessment.
- Sequential rationality is a refinement of Nash Equilibrium.
- The notion coincides with SPNE when applied to PIEFGs



Theorem

In a PIEFG, a behavioral strategy profile σ is an SPNE iff the tuple $(\sigma, \hat{\mu})$ is sequentially rational.

In a PIEFG, every information set is a singleton, $\hat{\mu}$ is the degenerate distribution at that singleton.

Equilibrium with Sequential rationality

Perfect Bayesian Equilibrium: An assessment (σ, μ) is PBE if $\forall i \in N$

- μ_i is Bayesian w.r.t. σ
 - σ_i is sequentially rational given σ_{-i} and μ_i
-
- Often represented only with σ , since μ is obtained from σ
 - Self-enforcing (like the SPNE) in a Bayesian way.

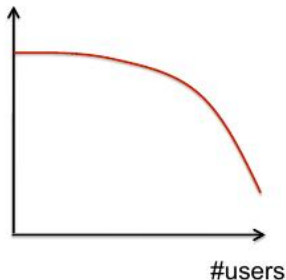


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Peer to Peer¹

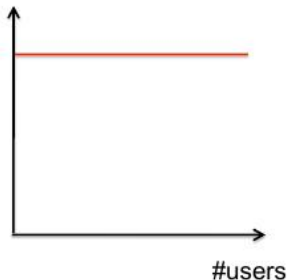


download rate



traditional

download rate



P2P

¹Slides of this section are adapted from CS186, Harvard

Desired Properties and Terminology



- Scalability
- Failure resilience

Terminology:

- **Protocol:** messages that can be sent, actions that can be taken over the network
- **Client:** a particular process for sending messages, taking actions
- **Reference client:** particular implementation
- **Peer**



Napster (1999 - 2001)

- Centralized database
- Users download music from each other

Gnutella (2000 -)

- Get list of IP addresses of peers from set of known peers (no server)
- To get a file: Query message broadcast by peer A to known peers
- Query response: sent by B if B has the desired file (routed back to requestor)
- A can then download directly from B

The File Sharing Game



Player 1	Player 2	
	Share	Free-ride
Share	2, 2	-1, 3, -1
Free-ride	0, 0	

(Gnutella) File Sharing Game

The File Sharing Game (Contd.)

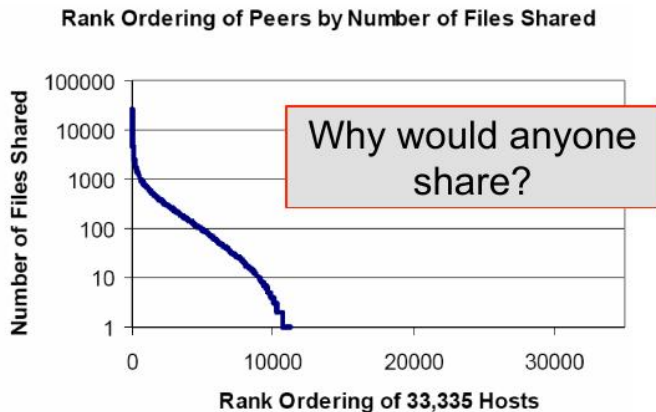


Image courtesy: Adar and Huberman (2000)

Incentives for Client Developers



- Client developers can ensure file sharing
- But competition among the developers
- 85% peers free-riding by 2005; Gnutella less than 1% of ww P2P traffic by 2013
- Few other P2P systems met the same fate



BitTorrent (2001 -)

- Approx 85% of P2P traffic in US
- File sharing
- Also used for S/W distribution (e.g., Linux)

Key innovations

- Break file into pieces: A repeated game!
- “If you let me download, I’ll reciprocate.”

BitTorrent Schematic

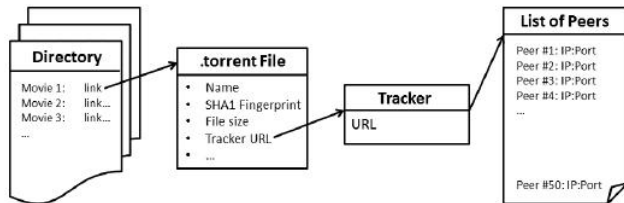


Figure 5.4.: Starting a download process in the BitTorrent protocol: 1) A user goes to a searchable directory to find a link to a .torrent file corresponding to the desired content; 2) the .torrent file contains metadata about the content, in particular the URL of a tracker; 3) the tracker provides a list of peers participating in the swarm for the content (i.e., their IP address and port); 4) the user's BitTorrent client can now contact all these peers and download content.

BitTorrent Optimistic Unchoking Algorithm



Tracker is a centralized entity that controls the traffic, tracks the connection between peers and their speed of upload, download etc.

Reference Client Protocol:

- Set a threshold r of uploading speed (typically the third maximum speed in the recent past)
- If a peer j uploaded to i at a rate $\geq r$, unchoke j in the next period
- If a peer j uploaded to i at a rate $< r$, choke j in the next period
- Every three time periods, optimistically unchoke a random peer from the neighborhood who is currently choked, and leave that peer unchoked for three time periods.

Forcing a repeated game by fragmenting the files

The leecher-seeder game is a repeated Prisoners' Dilemma

Strategy of the seeder is tit-for-tat



Illustration



- How often to contact tracker?
- Which pieces to reveal?
- How many upload slots, which peers to unchoke, at what speed?
- What data to allow others to download?
- Possible goals: min upload, max download speed, some balance

Attacks on BitTorrent



- BitThief
- Strategic piece revealer
- BitTyrant



- Goal: download files without uploading
- Keep asking for peers from tracker, grow neighborhood quickly
- Exploit the optimistic unchoking part
- Never upload!
- **Fix: modify the tracker (block same IP address within 30 minutes).**

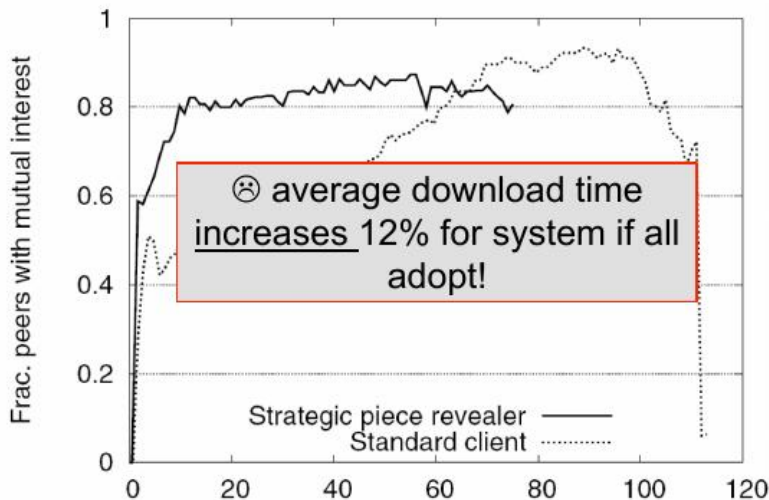
Ref: Locher et al., “Free Riding in BitTorrent is Cheap”, HotNets 2006



- Reference client: tell neighbors about new pieces, use “rarest-first” to request
- Manipulator strategy: reveal most common piece that reciprocating peer does not have!
- Try to protect a monopoly, keep others interested

Ref: Levin et al., “BitTorrent is an Auction: Analyzing and Improving BitTorrent’s Incentives”, SIGCOMM 2008

Strategic Piece Revealer





- P2P demonstrates importance of game-theory in computer systems
- Early systems were easily manipulated
- BitTorrent's innovation was to break files into pieces, enabling TitForTat.
- Still some vulnerabilities, but generally very successful example of incentive-based protocol design.



- ▶ Equilibrium in IIEFGs
- ▶ Game Theory in Practice: P2P File Sharing
- ▶ **Bayesian Games**
- ▶ Strategy, Utility in Bayesian Games
- ▶ Equilibrium in Bayesian Games
- ▶ Examples in Bayesian Equilibrium



Games

- **Non-cooperative games**
 - Complete information - Players **deterministically** know which game they are playing
 - Normal form games
Appropriate for simultaneous move single-stage games
Equilibrium notions: SDSE, WDSE, PSNE, MSNE, Correlated
 - Extensive form games
Appropriate for multi-stage games
Equilibrium notions: SPNE (PIEFG), mixed and behavioral strategies (IIEFG), PBE
 - Incomplete information - Players **do not deterministically** know which game they are playing
- Cooperative games - Players form coalitions and utilities are defined over coalitions
- Other types of games - repeated, stochastic etc.

Games with Incomplete Information



Games with Complete Information

- Players deterministically know the game they are playing
- There can be some chance moves but probabilities are known

Games with Incomplete information

- Players do not know deterministically know which game they are playing
- they receive **private signals / types**
- To discuss: a special subclass called games with incomplete information with **common priors** (Harsanyi 1967)
- Also called **Bayesian games**



Bayesian Games: Example

Football game (two competing teams)

- Each can choose a gameplan: aim to win (W) or aim to draw (D)
- We will call the gameplan as their **type** which are private signals to them, often caused by external factors, e.g., weather conditions, player injuries, ground conditions etc.
- There are four possible type profiles in this example WW, WD, DW, DD.
- The payoff matrices differ as follows (payoff for DW is symmetrically opposite to WD).

		FRA	
		ATT	DEF
ARG	ATT	1,1	2,0
	DEF	0,2	0,0

WW profile

		FRA	
		ATT	DEF
ARG	ATT	2,0	2,1
	DEF	0,1	1,0

WD profile

		FRA	
		ATT	DEF
ARG	ATT	0,0	1,0
	DEF	0,1	-1,-1

DD profile



Assumptions

- The probabilities of choosing different games (or type profiles) come from a **common prior** distribution.
- The common prior is common knowledge

Definition

A Bayesian game is represented by $\langle N, (\Theta_i)_{i \in N}, P, (\Gamma_\theta)_{\theta \in (\times_{i \in N} \Theta_i)} \rangle$

- N : set of players
- Θ_i : set of types of player i
- P : common prior distribution over $\Theta = \times_{i \in N} \Theta_i$
s.t. $\sum_{\theta_{-i} \in \Theta_{-i}} P(\theta_i, \theta_{-i}) > 0, \quad \forall \theta_i \in \Theta_i, \forall i \in N$
i.e., marginals for every type is positive (otherwise we can prune the type set)
- Γ_θ : NFG for the type profile $\theta \in \Theta$ i.e., $\Gamma_\theta = \langle N, (A_i(\theta))_{i \in N}, (u_i(\theta))_{i \in N} \rangle$
 $u_i : A \times \Theta \rightarrow \mathbb{R}, A = \times_{i \in N} A_i$ [We assume $A_i(\theta) = A_i, \forall \theta$]



Stages of a Bayesian game

- $\theta = (\theta_i, \theta_{-i})$ is chosen randomly according to the common prior P
- Each player observes her own type θ_i
- Player i picks action $a_i \in A_i, \forall i \in N$
- Player i realizes a payoff of $u_i(a_i, a_{-i}; \theta_i, \theta_{-i})$



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Definition

Strategy is a plan to map type to action.

$$s_i : \Theta_i \rightarrow A_i$$

$$\sigma_i : \Theta_i \rightarrow \Delta A_i$$

Pure

Mixed

The player can experience its utility in two stages for Bayesian games (depending on the realization of θ_i).

- Ex-ante utility
- Ex-interim utility
- Ex-post utility (for complete information game)



Definition (Ex-ante utility)

Expected utility before observing own type.

$$\begin{aligned} u_i(\sigma) &= \sum_{\theta \in \Theta} P(\theta) u_i(\sigma(\theta); \theta) \\ &= \sum_{\theta \in \Theta} P(\theta) \sum_{(a_1, a_2, \dots, a_n) \in A} \prod_{j \in N} \sigma_j(\theta_j) [a_j] u_i(a_1, \dots, a_n; \theta_1, \dots, \theta_n) \end{aligned}$$

The belief of player i over others' types changes after observing her own type θ_i according to Bayes rule on P .

$$P(\theta_{-i} | \theta_i) = \frac{P(\theta_i, \theta_{-i})}{\sum_{\tilde{\theta}_{-i} \in \Theta_{-i}} P(\theta_i, \tilde{\theta}_{-i})}$$

This is why we needed every marginal to be positive – otherwise that type can be removed from its type set



Definition (Ex-interim utility)

Expected utility after observing one's own type.

$$u_i(\sigma|\theta_i) = \sum_{\theta_{-i} \in \Theta_{-i}} P(\theta_{-i}|\theta_i) u_i(\sigma(\theta); \theta)$$

Special Case : for independent types, observing θ_i does not give any information on θ_{-i} . Both utilities are the same.

Relation between the two utilities is given by

$$u_i(\sigma) = \sum_{\theta_i \in \Theta_i} P(\theta_i) u_i(\sigma|\theta_i)$$



Example 1: Two Player Bargaining Game

- Player 1 : seller, type : price at which he is willing to sell
- Player 2 : buyer, type : price at which he is willing to buy
- $\Theta_1 = \Theta_2 = \{1, 2, \dots, 100\}, A_1 = A_2 = \{1, 2, \dots, 100\}$
- If the bid of the seller is smaller or equal to that of the buyer, trade happens at a price average of the two bids. Else, trade does not happen.



Example 1: Two Player Bargaining Game

Suppose type generation is independent and uniform over Θ_1, Θ_2 respectively,

$$P(\theta_2|\theta_1) = P(\theta_2) = \frac{1}{100}, \forall \theta_1, \theta_2$$

$$P(\theta_1|\theta_2) = P(\theta_1) = \frac{1}{100}, \forall \theta_1, \theta_2$$

$$u_1(a_1, a_2; \theta_1, \theta_2) = \begin{cases} \frac{a_1 + a_2}{2} - \theta_1 & \text{if } a_2 \geq a_1 \\ 0 & \text{otherwise} \end{cases}$$

$$u_2(a_1, a_2; \theta_1, \theta_2) = \begin{cases} \theta_2 - \frac{a_1 + a_2}{2} & \text{if } a_2 \geq a_1 \\ 0 & \text{otherwise} \end{cases}$$

Common Prior : $P(\theta_1, \theta_2) = \frac{1}{1000}, \forall \theta_1, \theta_2$



Example 2: Sealed Bid Auction

Two players, both willing to buy an object. Their values and bids lie in $[0,1]$.

Allocation Function:

$$O_1(b_1, b_2) = \begin{cases} 1 & \text{if } b_1 \geq b_2 \\ 0 & \text{ow} \end{cases} \quad O_2(b_1, b_2) = \begin{cases} 1 & \text{if } b_2 > b_1 \\ 0 & \text{ow} \end{cases}$$

Beliefs

$$f(\theta_2|\theta_1) = 1, \forall \theta_1, \theta_2$$

$$f(\theta_1|\theta_2) = 1, \forall \theta_1, \theta_2$$

$$f(\theta_1, \theta_2) = 1, \forall \theta_1, \theta_2$$

$$u_i(b_1, b_2; \theta_1, \theta_2) = O_i(b_1, b_2)(\theta_i - b_i)$$

Winner pays for his bid.



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Equilibrium concepts in Bayesian games

Ex-ante: before observing her own type

Nash Equilibrium (σ^*, P) : $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma'_i, \sigma_{-i}^*), \forall \sigma'_i, \forall i \in N$

$$u_i(\sigma) = \sum_{\theta \in \Theta} P(\theta) u_i(\sigma(\theta); \theta)$$

Ex-interim: after observing her own type

Bayesian Equilibrium (σ^*, P) : $u_i(\sigma_i^*(\theta_i), \sigma_{-i}^* | \theta_i) \geq u_i(\sigma'_i(\theta_i), \sigma_{-i}^* | \theta_i), \forall \sigma'_i, \forall \theta_i \in \Theta_i, \forall i \in N$

- The RHS of the definition can be replaced by a pure strategy $a_i, \forall a_i \in A_i$. The reason is exactly the same as that of MSNE (these definitions are equivalent)
- NE takes expectation over $P(\theta)$
- BE takes expectation over $P(\theta_{-i} | \theta_i)$

Equivalence of equilibrium concepts



Theorem

In finite Bayesian games, a strategy profile is Bayesian Equilibrium iff it is a Nash equilibrium

Proof.

For the forward direction, suppose (σ^*, P) is a Bayesian equilibrium, consider

$$\begin{aligned} u_i(\sigma'_i, \sigma_{-i}^*) &= \sum_{\theta_i \in \Theta_i} P(\theta_i) u_i(\sigma'_i(\theta_i), \sigma_{-i}^* | \theta_i) \\ &\leq \sum_{\theta_i \in \Theta_i} P(\theta_i) u_i(\sigma_i^*(\theta_i), \sigma_{-i}^* | \theta_i), \text{ since } (\sigma^*, P) \text{ is a BE} \\ &= u_i(\sigma_i^*, \sigma_{-i}^*) \end{aligned}$$





Equivalence of equilibrium concepts

Theorem

In finite Bayesian games, a strategy profile is Bayesian Equilibrium iff it is a Nash equilibrium

Proof.

For the reverse direction, proof by contradiction. Suppose (σ^*, P) is not a Bayesian equilibrium i.e., there exists some $i \in N$, some $\theta_i \in \Theta_i$, some $a_i \in A_i$, s.t.

$$u_i(a_i, \sigma_{-i}^* | \theta_i) > u_i(\sigma_i^*(\theta_i), \sigma_{-i}^* | \theta_i)$$

Construct the strategy $\hat{\sigma}_i$ s.t.,

$$\hat{\sigma}_i(\theta'_i) = \sigma_i^*(\theta'_i), \forall \theta'_i \in \Theta_i \setminus \{\theta_i\}$$

$$\hat{\sigma}_i(\theta_i)[a_i] = 1, \hat{\sigma}_i(\theta_i)[b_i] = 0, \forall b_i \in A_i \setminus \{a_i\}$$



Equivalence of equilibrium concepts



Theorem

In finite Bayesian games, a strategy profile is Bayesian Equilibrium iff it is a Nash equilibrium

Proof.

Reverse direction proof continued ...

$$\begin{aligned} u_i(\hat{\sigma}_i, \sigma_{-i}^*) &= \sum_{\tilde{\theta}_i \in \Theta_i} P(\tilde{\theta}_i) u_i(\hat{\sigma}_i(\tilde{\theta}_i), \sigma_{-i}^* | \tilde{\theta}_i) \\ &= \sum_{\tilde{\theta}_i \in \Theta_i \setminus \{\theta_i\}} P(\tilde{\theta}_i) u_i(\hat{\sigma}_i(\tilde{\theta}_i), \sigma_{-i}^* | \tilde{\theta}_i) + P(\theta_i) u_i(\hat{\sigma}_i(\theta_i), \sigma_{-i}^* | \theta_i) \\ &> \sum_{\tilde{\theta}_i \in \Theta_i \setminus \{\theta_i\}} P(\tilde{\theta}_i) u_i(\sigma_i^*(\tilde{\theta}_i), \sigma_{-i}^* | \tilde{\theta}_i) + P(\theta_i) u_i(\sigma_i^*(\theta_i), \sigma_{-i}^* | \theta_i) = u_i(\sigma_i^*, \sigma_{-i}^*) \end{aligned}$$

Hence, $(\sigma_i^*, \sigma_{-i}^*)$ is not a Nash equilibrium



Existence of Bayesian Equilibrium



Theorem

Every finite Bayesian game has a Bayesian equilibrium.

[Finite Bayesian game: set of players, action set and type set are finite]

Proof.

Proof idea: Transform the Bayesian game into a complete information game treating each type as a player, and invoke Nash Theorem for the existence of equilibrium - which is a BE in the original game. [See addendum for details] □



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Allocation Function

$$O_1(b_1, b_2) = I\{b_1 \geq b_2\}$$

$$O_2(b_1, b_2) = I\{b_2 > b_1\}$$

Beliefs

$$f(\theta_2|\theta_1) = 1, \forall \theta_1, \theta_2$$

$$f(\theta_1|\theta_2) = 1, \forall \theta_1, \theta_2$$

$$f(\theta_1, \theta_2) = 1, \forall \theta_1, \theta_2$$

First Price Auction



- If $b_1 \geq b_2$ payer 1 wins and pays her bid otherwise, player 2 wins and pays her bid.

$$u_1(b_1, b_2, \theta_1, \theta_2) = (\theta_1 - b_1)T\{b_1 \geq b_2\}$$

$$u_2(b_1, b_2, \theta_1, \theta_2) = (\theta_2 - b_2)T\{b_1 < b_2\}$$

- $b_1 = s_1(\theta_1), b_2 = s_2(\theta_2)$
Assume $s_i(\theta_i) = \alpha_i \theta_i, \alpha_i > 0, i = 1, 2$

First Price Auction



To find the BE, we need to find the s_i^* (or α_i^*) that maximizes the ex-interim utility of player i . i.e.

$$\max_{\sigma_i} u_i(\sigma_i, \sigma_{-i}^* | \theta_i)$$

For player 1, this reduces to

$$\begin{aligned} \max_{\sigma_i} u_i(\sigma_i, \sigma_{-i}^* | \theta_i) &= \max_{b_1 \in [0, \alpha_2]} \int_0^1 f(\theta_2 | \theta_1) (\theta_1 - b_1) I\{b_1 \geq \alpha_2 \theta_2\} d\theta_2 \\ &= \max_{b_1 \in [0, \alpha_2]} (\theta_1 - b_1) \frac{b_1}{\alpha_2} \\ \implies b_1 &= \begin{cases} \frac{\theta_1}{2} & \text{if } \alpha_2 > \frac{\theta_1}{2} \\ \alpha_2 & \text{otherwise} \end{cases} \end{aligned}$$

First Price Auction



From this we get,

$$s_1^*(\theta_1) = \min\left\{\frac{\theta_1}{2}, \alpha_2\right\}$$

$$s_2^*(\theta_2) = \min\left\{\frac{\theta_2}{2}, \alpha_1\right\}$$

If $\alpha_1 = \alpha_2 = \frac{1}{2}$, then $(\frac{\theta_1}{2}, \frac{\theta_2}{2})$ is a BE.

In the Bayesian Game induced by uniform prior on first price auction, bidding half the true value is a Bayesian equilibrium.

Second Price Auction



Highest bidder wins but pays the second highest bid.

$$u_1(b_1, b_2, \theta_1, \theta_2) = (\theta_1 - b_2)T\{b_1 \geq b_2\}$$

$$u_2(b_1, b_2, \theta_1, \theta_2) = (\theta_2 - b_1)T\{b_1 < b_2\}$$

Player 1 has to maximize

$$\begin{aligned} &= \int_0^1 f(\theta_2|\theta_1)(\theta_1 - s_2(\theta_2))I\{b_1 \geq s_2(\theta_2)\}d\theta_2 \\ &= \int_0^1 1 \cdot (\theta_1 - \alpha_2\theta_2)I\{\theta_2 \leq \frac{b_1}{\alpha_2}\}d\theta_2 \\ &= \frac{1}{\alpha_2}(b_1\theta_1 - \frac{\theta_1^2}{2}) \end{aligned}$$

This is maximized when $b_1 = \theta_1$. Similarly for $b_2 = \theta_2$.



Second Price Auction

If the distribution of θ_1 and θ_2 were arbitrary but independent, the maximization problem would have been

$$\int_0^{\frac{b_1}{\alpha_2}} f(\theta_2)(\theta_1 - \alpha_2 \theta_2) d\theta_2 = \theta_1 F\left(\frac{b_1}{\alpha_2}\right) - \alpha_2 \int_0^{\frac{b_1}{\alpha_2}} \theta_2 f(\theta_2) d\theta_2$$

Differentiating wrt b_1 , we get

$$\theta_1 \frac{1}{\alpha_2} f\left(\frac{b_1}{\alpha_2}\right) - \alpha_2 \cdot \frac{b_1}{\alpha_2} f\left(\frac{b_1}{\alpha_2}\right) \frac{1}{\alpha_2} = 0 \implies \frac{1}{\alpha_2} f\left(\frac{b_1}{\alpha_2}(b_1 - \theta_1)\right) = 0 \quad (1)$$

$$\implies b_1 = \theta_1 \text{ if } f\left(\frac{b_1}{\alpha_2}\right) > 0 \quad (2)$$

Similarly for 2.

For any independent positive prior, bidding true type is a BE of the induced Bayesian game in Second Price Auction.



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